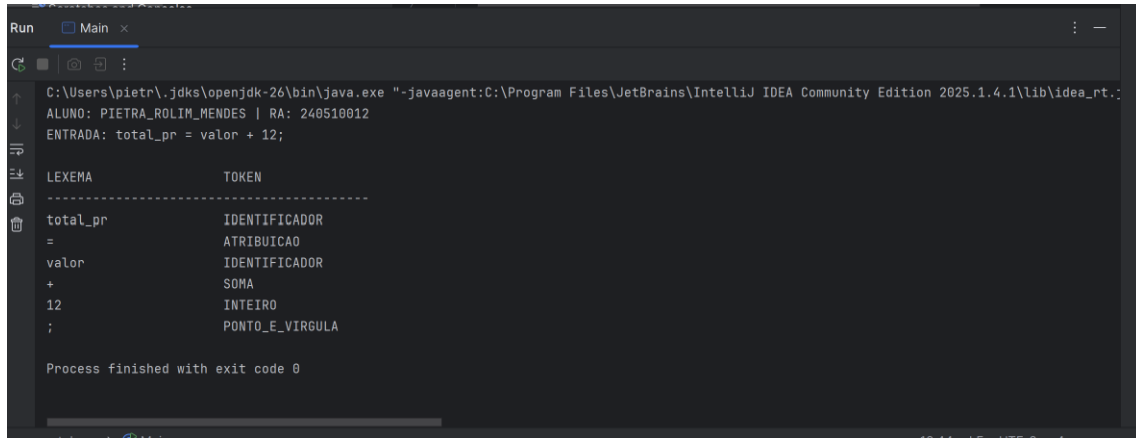


PRINTS ATIVIDADE AULA EXPLORATORIA CAPITULO 4:

01_tokens:



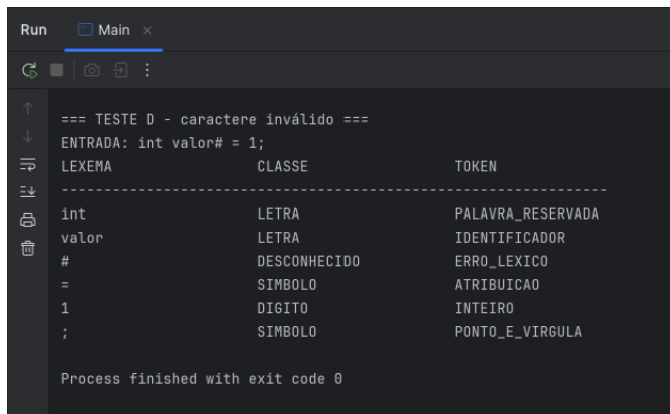
The screenshot shows the Run window of an IDE. The command line at the top indicates the Java executable and the Java agent path. Below the command line, the program output is displayed. It starts with the student information: "ALUNO: PIETRA_ROLIM_MENDES | RA: 240510012" and the input statement: "ENTRADA: total_pr = valor + 12;". This is followed by a table showing the tokens of the input statement. The table has two columns: "LEXEMA" and "TOKEN". The tokens are: "total_pr" (IDENTIFICADOR), "=" (ATRIBUICAO), "valor" (IDENTIFICADOR), "+" (SOMA), "12" (INTEIRO), and ";" (PONTO_E_VIRGULA). The output ends with "Process finished with exit code 0".

```
C:\Users\pietr\.jdk\openjdk-26\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1.4.1\lib\idea_rt.jar:..."
ALUNO: PIETRA_ROLIM_MENDES | RA: 240510012
ENTRADA: total_pr = valor + 12;

LEXEMA          TOKEN
-----
total_pr        IDENTIFICADOR
=               ATRIBUICAO
valor           IDENTIFICADOR
+              SOMA
12              INTEIRO
;               PONTO_E_VIRGULA

Process finished with exit code 0
```

02_lexico:



The screenshot shows the Run window of an IDE. The command line at the top indicates the Java executable and the Java agent path. Below the command line, the program output is displayed. It starts with the student information: "ALUNO: PIETRA_ROLIM_MENDES | RA: 240510012" and the input statement: "ENTRADA: int valor# = 1;". This is followed by a table showing the tokens of the input statement. The table has three columns: "LEXEMA", "CLASSE", and "TOKEN". The tokens are: "int" (PALAVRA_RESERVADA), "valor" (IDENTIFICADOR), "#" (ERRO_LEXICO), "=" (ATRIBUICAO), "1" (INTEIRO), and ";" (PONTO_E_VIRGULA). The output ends with "Process finished with exit code 0".

```
=== TESTE 0 - caractere inválido ===
ALUNO: PIETRA_ROLIM_MENDES | RA: 240510012
ENTRADA: int valor# = 1;

LEXEMA          CLASSE          TOKEN
-----
int             PALAVRA_RESERVADA
valor           IDENTIFICADOR
#               ERRO_LEXICO
=               ATRIBUICAO
1               INTEIRO
;               PONTO_E_VIRGULA

Process finished with exit code 0
```

03_erro_compilador:



The screenshot shows the Build window of an IDE. The command line at the top indicates the Java executable and the Java agent path. Below the command line, the program output is displayed. It starts with the student information: "ALUNO: PIETRA_ROLIM_MENDES | RA: 240510012" and the input statement: "ENTRADA: int valor# = 1;". This is followed by a table showing the tokens of the input statement. The table has three columns: "LEXEMA", "CLASSE", and "TOKEN". The tokens are: "int" (PALAVRA_RESERVADA), "valor" (IDENTIFICADOR), "#" (ERRO_LEXICO), "=" (ATRIBUICAO), "1" (INTEIRO), and ";" (PONTO_E_VIRGULA). The output ends with "Process finished with exit code 0".

```
C:\Users\pietr\Downloads\Programas_Java_Capitulo_4\Programas_Java_Capitulo_4\03-diagnostics\Main.java:8:28
java: ';' expected
```

```
Run Main x
C:\Users\pietr\.jdk\openjdk-26\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1.4.1\lib\idea_rt...
ALUNO: PIETRA_ROLIM_MENDES | RA: 240510012
Programa corrigido!
Resultado: 30
Process finished with exit code 0
```

04_descendente:

```
Run Main x
C:\Users\pietr\.jdk\openjdk-26\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1.4.1\lib\idea_rt...
ALUNO: PIETRA_ROLIM_MENDES | RA: 240510012
ENTRADA: (2 + 3) * 4
ENTRANDO em <expr> | lookahead = (
  ENTRANDO em <term> | lookahead = (
    ENTRANDO em <factor> | lookahead = (
      Abrindo parenteses
      Novo lookahead: 2
    ENTRANDO em <expr> | lookahead = 2
      ENTRANDO em <term> | lookahead = 2
        ENTRANDO em <factor> | lookahead = 2
          Numero reconhecido: 2
          Novo lookahead: +
        SAINDO de <factor> | lookahead = +
      SAINDO de <term> | lookahead = +
    Encontrado operador +
    Novo lookahead: 3
  ENTRANDO em <term> | lookahead = 3
    ENTRANDO em <factor> | lookahead = 3
      Numero reconhecido: 3
      Novo lookahead: )
    )
  )
)
```

```
    Novo lookahead: )
  SAINDO de <factor> | lookahead = )
  SAINDO de <term> | lookahead = )
  SAINDO de <expr> | lookahead = )
  Novo lookahead: *
  Fechando parenteses
  SAINDO de <factor> | lookahead = *
  Encontrado operador *
  Novo lookahead: 4
  ENTRANDO em <factor> | lookahead = 4
    Numero reconhecido: 4
    Novo lookahead: $FIM$
  SAINDO de <factor> | lookahead = $FIM$
  SAINDO de <term> | lookahead = $FIM$
  SAINDO de <expr> | lookahead = $FIM$
```

```
ENTRADA ACEITA
RESULTADO DA EXPRESSAO: 20
```

```
Process finished with exit code 0
```

05_ascendente:

RunMain

C:\Users\pietr\.jdk\openjdk-26\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1.4.1\lib\idea_rt...

ALUNO: PIETRA_ROLIM_MENDE | RA: 240510012

ENTRADA: id+id*id

PILHA	ENTRADA	ACAO
\$ 0	id+id*id\$	SHIFT id -> estado 5
\$ 0 id 5	+id*id\$	REDUCE F -> id
\$ 0 F 3	+id*id\$	REDUCE T -> F
\$ 0 T 2	+id*id\$	REDUCE E -> T
\$ 0 E 1	+id*id\$	SHIFT + -> estado 6
\$ 0 E 1 + 6	id*id\$	SHIFT id -> estado 5
\$ 0 E 1 + 6 id 5	*id\$	REDUCE F -> id
\$ 0 E 1 + 6 F 3	*id\$	REDUCE T -> F
\$ 0 E 1 + 6 T 9	*id\$	SHIFT * -> estado 7
\$ 0 E 1 + 6 T 9 * 7	id\$	SHIFT id -> estado 5
\$ 0 E 1 + 6 T 9 * 7 id 5	\$	REDUCE F -> id
\$ 0 E 1 + 6 T 9 * 7 F 10	\$	REDUCE T -> T * F
\$ 0 E 1 + 6 T 9	\$	REDUCE E -> E + T
\$ 0 E 1	\$	ACCEPT

Process finished with exit code 0